20226번 - Luggage 성공 다국어

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4 다이아몬드 IV

시간 제한	메모리 제한	제출	정답	맞힌 사람	정답 비율
10 초 (추가 시간 없음)	512 MB	823	229	114	22.441%

문제

Mr. Yokohama is estimating the delivery fare for a number of luggage pieces. The fare for a piece is determined by its width *w*, depth *d* and height *h*, each rounded up to an integer. The fare is proportional to the *sum* of them.

The list of the luggage pieces is at hand, but the list has the *product* $w \times d \times h$ for each piece, instead of the *sum* $w+d+h$ by mistake. This information is not enough to calculate the exact delivery fare.

Mr. Yokohama therefore decided to estimate the minimum possible delivery fare. To this end, given the listed *product* p for each of the luggage pieces, the minimum possible *sum* $s=w+d+h$ satisfying $p=w \times d \times h$ should be found.

You are requested to help Mr. Yokohama by writing a program that, when given the product p, computes the minimum sum s.

입력

The input consists of multiple datasets. Each of the datasets has one line containing an integer p ($0 < p < 10^{15}$).

The end of the input is indicated by a line containing a zero.

The number of datasets does not exceed 300.

출력

For each dataset, output a single line containing an integer s. s should be the minimum possible sum $w+d+h$ of three positive integers, w, d, and h, satisfying $p=w \times d \times h$.

예제 입력 1 복사

```
1
2
6
8
729
47045881
12137858827249
562949953421312
986387345919360
999730024299271
999998765536093
0
```

예제 출력 1 복사

```
3
4
6
6
27
1083
6967887
262144
298633
299973
999998765536095
```

출처

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메모